

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2168

Waukena 138 kV
September 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2168. The Interconnection Customer proposes a 70.6 MW solar photovoltaic facility interconnecting at the Waukena 138 kV substation in the Pennsylvania area. The request was received on August 28, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 708 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,067,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Marquis – Friendly Hills 138 kV	Ritter Park – Sugar Hill 345 kV line	154.2	254	6.2
Guntersville Dam – Sugar Hill 138 kV Line	Sugar Hill – Ritter Park 345 kV	153.9	217	6.2
Belfield – Guntersville Dam 138 kV	Sugar Hill – Ritter Park 345 kV	149.9	241	6.2
Friendly Hills – Uc Davis 138 kV L	Ritter Park – Sugar Hill 345 kV	146.8	278	6.2
Marquis – Belfield 138 kV	Sugar Hill – Ritter Park Substation 345 kV	144.0	261	6.2
Uc Davis – Ritter Park 138 kV	Sugar Hill – Ritter Park 345 kV	136.2	312	6.2
Stewart Mn Dam – Shepaug 138 kV Line	Ritter Park – Sugar Hill Substation 345 kV	126.6	390	10.8
N – Shepaug 138 kV L	Sugar Hill – Ritter Park 345 kV Line	112.6	429	10.8
Ritter Park – Bullard 138 kV	Stewart Mtn Dam – Ritter Park 345 kV Line	103.8	305	20.1
Toston – Bullard 138 kV	Ritter Park – Stewart Mtn Dam 345 kV	103.1	322	20.1
Torrey Substation – Toston 138 kV	Stewart Mountain Dam – Ritter Park 345 kV L	102.6	337	20.1
Blountsville – Vista 138 kV	Stewart Mountain Dam 345 kV/138 kV Auto	102.4	208	15.3
Blountsville Substation – Torrey 138 kV Line	Stewart Mtn Dam – Ritter Park 345 kV	102.1	352	20.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Friendly Hills – Marquis 138 kV	Rebuild line with higher-capacity conductor	\$3,232,000	35% (\$1,139,000)
Guntersville Dam – Sugar Hill 138 kV Line	Rebuild line with higher-capacity conductor	\$3,080,000	35% (\$1,086,000)
Belfield – Guntersville Dam 138 kV line	Rebuild line with higher-capacity conductor	\$2,294,000	35% (\$809,000)
Friendly Hills – Uc Davis 138 kV L	Rebuild line with higher-capacity conductor	\$2,425,000	35% (\$855,000)
Marquis – Belfield 138 kV	Rebuild line with higher-capacity conductor	\$3,375,000	35% (\$1,190,000)
Uc Davis – Ritter Park 138 kV line	Rebuild line with higher-capacity conductor	\$1,721,000	35% (\$607,000)
Stewart Mn Dam – Shepaug 138 kV Line	Rebuild line with higher-capacity conductor	\$994,000	15% (\$146,000)
N – Shepaug 138 kV L	Rebuild line with higher-capacity conductor	\$5,982,000	15% (\$879,000)
Ritter Park – Bullard 138 kV	Reconductor line	\$306,000	66% (\$203,000)
Toston – Bullard 138 kV	Reconductor line	\$1,592,000	66% (\$1,056,000)
Toson – Torrey 138 kV	Reconductor line	\$465,000	66% (\$308,000)
Blountsville – Vista 138 kV	Reconductor line	\$552,000	42% (\$234,000)
Blontsville – Torrey Substation 138 kV L	Reconductor line	\$836,000	66% (\$555,000)
	Total allocated to project		\$9,067,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.